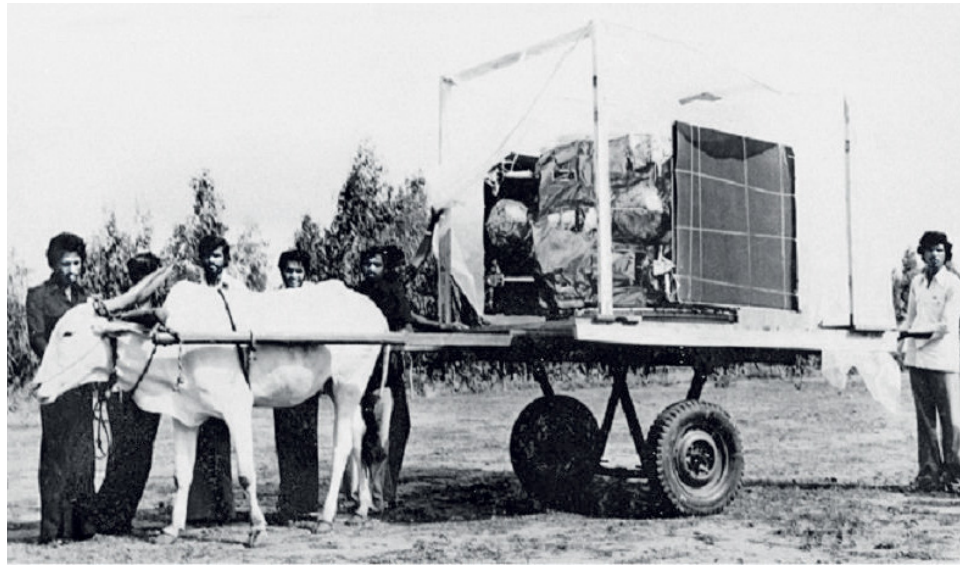
STELLAR SHOW

The Indian space programme is now among the world's most accomplished, instilling pride in every Indian

By ***Amarnath K. Menon***



SPACE CRAFTSMANSHIP Left, ISRO scientists carry India's first communication satellite on a bullock cart in 1981; right, a truck carries part of the GSLV Mk III rocket for the launch of ISRO's GSAT29 satellite in 2018

IN *Indian public consciousness,* the stupendous progress in space technology often stands in for the country's overall advancement in science. Ever since India launched the first US-made sounding rocket for upper atmospheric research on November 21, 1963, its ambitious space programme has not looked back. This came a year after Vikram Sarabhai formed the Indian National Committee for Space Research (INCOSPAR), which became the Indian Space Research Organisation (ISRO) on August 15, 1969.

Soon, it sent India's first satellite, Aryabhata, into space in 1975. India's entry into the elite space club came in 1980, with the launch of the satellite launch vehicle SLV-3. Since then, there has been steady progress in developing heavy launchers and satellites. The Polar Satellite Launch Vehicle (PSLV) and the Geostationary Satellite Launch Vehicle (GSLV) are workhorses in lobbing Indian and foreign satellites into space. ISRO has also provided communication links using INSAT satellites and deployed remote-sensing satellites to map India's vast resources. By the 2000s, it had developed capabilities for space exploration. Chandrayaan-1, the unmanned lunar space probe, was a technological leap and provided impetus for the Mars Orbiter Mission, India's first interplanetary mission. Mangalyaan, was launched in November 2013 and reached the Mars orbit in September 2014, making India the first nation to achieve this in its maiden attempt. While Rakesh Sharma was the first Indian in space in 1984 aboard the Soyuz T-11, ISRO aims to send an Indian to space in an Indian spacecraft. ■

A.P.J. ABDUL KALAM, WHO WAS ON INCOSPAR'S TEAM OF ROCKET ENGINEERS AND DIRECTOR OF THE PIONEERING SLV-3 MISSION, LATER BECAME PRESIDENT OF INDIA

IN 2017, THE PSLV-C37 ROCKET LAUNCHED 104 SATELLITES INTO ORBIT IN A SINGLE FLIGHT–A WORLD RECORD